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Article in *Technology and Disability* · June 2020

DOI: 10.3233/TAD-200262

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Assessing postoperative cognitive dysfunction using 3D multiple object tracking in open heart surgery patients

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Abstract.

BACKGROUND: Post-operative cognitive dysfunction is a common complication after heart surgery that affects up to 60% of all open-heart surgery patients. Despite its prevalence, limited attention has been given to different methods to retrain cognition in open-heart surgery patients.

OBJECTIVE: To examine whether 3-dimensional multiple object tracking (3D MOT) can be used to detect changes in cognitive function in open-heart surgery patients.

METHODS: In total, 16 open-heart surgery patients (age: 59.43 ± 12.99 years) from a Midwestern Canadian hospital were recruited. The patients completed a cognitive assessment, including 3D MOT and other standardized neurocognitive tests at 3 time points: 1 to 2 days pre-surgery, at discharge or 1-week post-surgery (whichever came first), and at 12-weeks post-surgery.

RESULTS: No significant differences were detected between baseline and 1-week/discharge measurements on all measures. Patients improved significantly from 1-week/discharge to 12-weeks in 3D MOT scores. A similar yet non-significant ($p = 0.07$) trend was found on some neurocognitive tests (i.e., Montreal Cognitive Assessment).

CONCLUSION: No significant decline from pre- to 1-week/discharge post-surgery was found on all measures. 3D MOT detected post-surgical cognitive changes in open-heart surgery patients. Future research is warranted to explore the potential of 3D MOT in retraining cognition after heart surgery.

Keywords: POCD, cognitive rehabilitation, retraining, CABG, AVR

1. Introduction

Post-operative cognitive dysfunction (POCD), defined new cognitive deficits that occur post-operatively [1], is a well-recognized temporary post-surgical complication that affects up to 60% of patients

undergoing coronary artery bypass graft (CABG) or aortic valve replacement (AVR) surgery [2]. As such, it can only be tested by using pre- and post-surgical assessment of cognitive function. Various cognitive domains may be affected for several weeks but patients exhibiting POCD will eventually recover from the cognitive impairments [3]. Yet, POCD has been associated with severe impairments of quality of life and greater utilization of health care resources during the time of post-surgical cognitive impairment [4].

Several risk factors for POCD have been identi-

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fied, including individual (e.g., pre-existing cerebrovascular disease, genetic predisposition) and intraoperative (e.g., type of surgery, cerebral embolism) factors [5]. However, two of the most prevalent risk factors of POCD are the presence of pre-surgical cognitive dysfunction and age [6]. To date, several pharmacologic interventions have been trialed to counteract the acute effects of POCD. Unfortunately, most of the current trials showed little effect to improve cognition [5].

Most of the current research has focused on establishing diagnostic criteria and the prevalence of POCD. Less attention has been given to non-invasive, non-pharmaceutical methods of retraining cognition after heart surgery. This is surprising as cognitive retraining has been a well-established rehabilitation method to enhance quality of life in other cognitively impaired patients (e.g., Multiple Sclerosis) [7]. The only study of cognitive retraining in CABG patients indicated significant improvements in attention and memory of trained CABG patients compared to a control group [8]. The study revealed that cognitive training may be a practical mode of cognitive rehabilitation in this population. Yet, more research into the effectiveness of cognitive training is clearly needed.

Recently, cognitive training software, called Neurotracker (Cognisens, Montreal, QC, Canada), has been developed, which targets selective, sustained, and dynamic attention as well as visual processing speed. All of these cognitive domains may be impaired in POCD patients. The software uses 3D Multiple Object Tracking (3D MOT). Patients have to track spheres in a three-dimensional space, while ignoring decoys. The advantage of using 3D MOT include [9]: 1) a large visual field and binocular 3D, which are required in activities of daily living 2) adaptive workloads to optimally improve cognitive function, and 3) a computerized task, which is easily accessible even from remote locations. Initial validation studies showed measurable improvements in attention, processing speed, and working memory as well as changes in neuroelectric brain function at rest [10]. Several studies have linked 3D MOT to enhanced cognitive function in elderly populations [11,12] and transferability to real-life activities, such as driving [13] and other performance domains (e.g., athletics [14,15], laparoscopic surgery [16]). In clinical populations, 3D MOT has been used in children with cognitive disorders (e.g., autism spectrum disorder, ADHD) [17] and patients with Multiple Sclerosis [18]. The current evidence indicated substantial training effects of the intervention and some transfer effects to other cognitive tests (e.g.,

Useful Field of View [18]) in clinical and non-clinical populations, supporting the validity of 3D MOT training for the enhancement of attention. Despite the promising evidence from existing research, the effectiveness of 3D MOT to assess and/or train cognitive functioning has yet to be tested in open-heart surgery patients.

The first step in addressing this shortcoming is to test if 3D MOT is sensitive to cognitive changes that are associated with POCD. Hence, the primary purpose of the present pilot study is to evaluate whether 3D MOT can detect cognitive changes from pre- to post- (i.e., discharge/1 week and 12 week) heart surgery. There are two hypotheses: First, it was assumed that POCD patients would show a significant decline in 3D MOT performance from pre-surgery to one week after surgery (or discharge from hospital). Second, it was hypothesized that patients would significantly improve their 3D MOT performance from 1 week (or discharge from hospital) to the 12-week post-surgery. The secondary purpose was to test any changes in cognition over time with other validated measures (i.e., the Montreal Cognitive Assessment [MOCA], the Trails B Test).

2. Materials and methods

2.1. Design

The present study employed an observational, uncontrolled pilot study design.

2.2. Participants

Sixteen open-heart surgery patients (3 female, 18.8%, mean age = 59.44 ± 12.99 years, age range = 22–80) admitted to the General Hospital in Regina, SK, Canada, were enrolled in the study. The patients underwent AVR ($n = 5$), CABG ($n = 2$), AVR and CABG ($n = 2$), or other open-heart surgery ($n = 2$). All surgeries were performed in the Regina General Hospital using the same heart-lung machine and perfusion guidelines. Data collection occurred between April 2016 and May 2017. Patients were excluded if they a) needed to receive emergent surgery, b) had a previous surgery in the past 12 months, c) underwent previous open-heart surgery, d) had a diagnosis of cognitive impairment (e.g., dementia), e) were unable to read Times New Roman at a font size of 16 without glasses, or f) were unable to provide consent to the study for themselves. Prior to enrollment, the study was approved by the health region's ethics board and all participants signed an informed consent form.

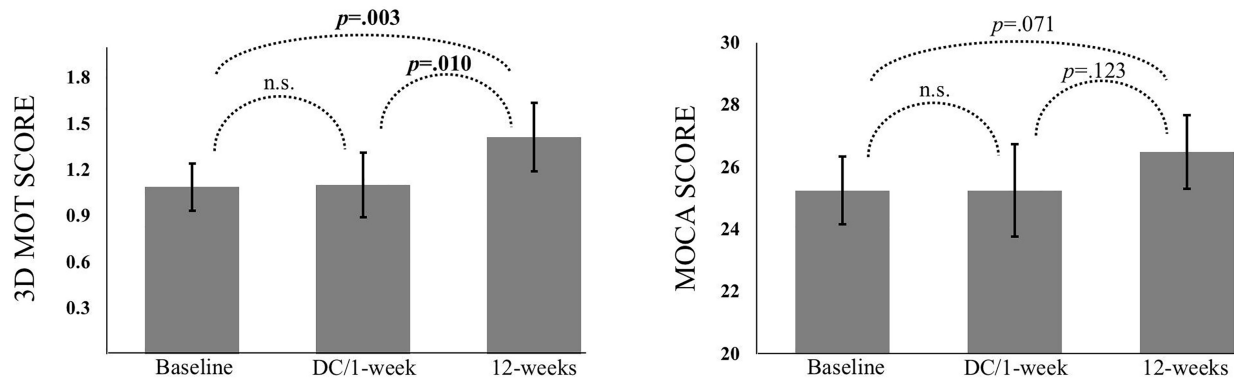


Fig. 1. Comparison of 3D MOT and MOCA score over study timepoints.

2.3. Study protocol

For all patients, the baseline measures were taken in hospital between 24 and 48 hours before the heart surgery. Patients were contacted after 1 week of hospital stay had elapsed (to eliminate the potential effects of anesthesia on cognition) or upon discharge, whichever occurred first. After 12 weeks, patients were contacted again. A research assistant travelled to a convenient location for the patient and administered the final wave of data collection. For all tests (i.e., baseline, discharge/1 week, and 12 week), the same test battery and procedures were used.

2.4. Measurements

2.4.1. 3D MOT

Multiple object tracking was assessed using the Neurotracker® software. Participants wore a head mounted display (Sony, Inc.) that enabled standardized conditions for testing. The software projected eight spheres into a 3D space (see Fig. 1) onto the goggle screen; three spheres were illuminated for two seconds by red halos. The halos then disappeared and the balls moved around randomly in the space for eight seconds. After eight seconds, the spheres were numbered and the participant attempted to identify the three objects that were initially highlighted by a halo.

The speed of the objects' movement increased or decreased with each trial based on the participant's accuracy. All three objects must have been correctly identified to increase the speed of the next trial. One session comprised 20 repetitions of this procedure and took approximately seven minutes to complete. The software calculated an average of the normalized speed threshold based on the accuracy of object identification for each session of 20 repetitions. Three sessions were completed at each assessment. The average score of all three sessions was the outcome variable.

2.4.2. Neurocognitive battery

The selection of neurocognitive tests was based on recommendation by Ghoneim and Block [5].

2.4.2.1. Montreal cognitive assessment (MOCA) [19]

The MOCA is a brief, paper-based test designed to assess cognitive skills including visuospatial abilities, memory, executive function, attention, concentration, working memory, language and orientation. There is substantial evidence that the MOCA is sensitive to detecting cognitive impairment in various clinical populations, including POCD patients.

2.4.2.2. Trails B [20]

The Trails B is one of the most popular neuropsychological tests and is included in most batteries to assess POCD. The test provides information on visual search speed, processing speed, mental flexibility, and executive function. The test requires participants to correctly connect a sequence of numbers and letters (i.e., 1, A, 2, B, 3, C ...) as quickly as possible. The time in seconds was recorded for each trial.

2.5. Statistical analysis

All variables were sufficiently distributed (Skewness < 3, Kurtosis < 7) for the calculation of parametric statistics. To compare changes in the dependent variables (i.e., 3DMOT speed, MOCA, Trails B) over the three time points (baseline, discharge/1 week, 12 weeks), three repeated measures analysis of variance with Omega squared effect sizes were conducted. To compare differences between time points, Bonferroni post hoc analyses with Cohen's d effect sizes were calculated. An alpha level was set at 0.05 for the present study. All analyses were conducted in JASP 0.9.0.1 and SPSS 25 (IBM).

Table 1
Patient characteristics and descriptives

Variable	Mean $\pm$ SD/Count %	Median	Range
N	16		
Sex	Male 13, 81.25% Female 3, 18.75%		
Age (years)	59.44 $\pm$ 12.99	59.5	22–80
Baseline 3D MOT	1.09 $\pm$ 0.29	0.96	0.72–1.54
DC/1 week 3D MOT	1.10 $\pm$ 0.39	1.11	0.41–1.77
12 weeks 3D MOT	1.41 $\pm$ 0.42	1.38	0.78–2.35
Baseline MOCA	25.25 $\pm$ 2.05	25	21–29
DC/1 week MOCA	25.25 $\pm$ 2.80	25.50	20–30
12 weeks MOCA	26.50 $\pm$ 2.22	26.50	22–30
Baseline Trails B	72.00 $\pm$ 35.12	66.50	32–161
DC/1 week Trails B	76.25 $\pm$ 42.34	61	25–181
12 weeks Trails B	65.69 $\pm$ 29.03	58.50	25–118

3. Results

3.1. 3D MOT

A significant improvement in 3D MOT speed threshold was found ($F_{2,30} = 9.56$, $p < 0.001$, $\omega^2 = 0.13$). Bonferroni post hoc analyses revealed no statistically significant difference between baseline and discharge/1 week measures ($t = -0.18$, $p = 1.00$). However, significant differences existed between baseline and 12 weeks ($t = -4.13$, $p = 0.003$, $d = -1.03$) and discharge/1 week and 12 weeks ($t = -3.48$, $p = 0.01$, $d = -0.87$). Results indicate that patients maintained their 3D MOT performance from baseline to discharge/1 week. However, the 3D MOT performance significantly increased 12 weeks after discharge.

3.2. MOCA

A significant difference in MOCA scores were found ($F_{2,30} = 4.03$, $p = 0.03$, $\omega^2 = 0.04$). Bonferroni post hoc analyses revealed no statistically significant difference between baseline and discharge/1 week measures ($t = 0.00$, $p = 1.00$). However, the differences between baseline and 12 weeks ($t = -2.52$, $p = 0.07$, $d = -0.63$) and discharge/1 week and 12 weeks ($t = -2.24$, $p = 0.12$, $d = -0.56$) approached significance. Similar to 3DMOT, no significant differences were found between baseline and discharge/1 week. While not significant, a trend in the data suggests that patients scored higher after 12 weeks.

3.3. Trails B

No significant difference across the time points for Trails B performance was found ($F_{1,29,19.41} = 1.42$, $p = 0.26$).

4. Discussion

The present study showed that 3D MOT detected a significant improvement in cognitive function 12 weeks after open-heart surgery. Similar trends were observable in another neurocognitive assessment (i.e., MOCA).

4.1. Assessment of POCD using 3D MOT

Contrary to the first hypothesis, no significant decrease in 3D MOT performance between baseline and discharge/1 week was observed. There are several possible explanations for this finding. For organizational reasons, the patients were assessed 24–48 hours before the surgery. The anticipation of the surgery may negatively influence several cognitive domains [21]. In addition, the patients' baseline performance may have been lower than their true cognitive potential. Second, the patients were relatively young and comorbidity-free to eliminate dementia as a potential confounding factor. Several reviews have suggested that age and pre-operative cognitive decline, among other factors, are influential predictors of POCD [5,6]. As such, the patients in this study might not have experienced substantial cognitive decline as a result of the surgery. The study should be replicated with a more diverse patient population and earlier (or multiple) baseline measurements. Yet, the data supported our second hypothesis. Patients increased their 3D MOT performance significantly from discharge/1 week to 12 weeks post-surgery. This indicated that 3D MOT might be usable to detect post-surgical cognitive changes in open-heart surgery patients.

4.2. Changes in MOCA and Trails B

There were no significant changes in cognition de-

tected by the MOCA and Trails B. However, the trends in the MOCA data indicated that patients retained similar cognitive function from baseline to discharge/1 week, while almost showing improvements in cognition from baseline to 12 weeks ($p = 0.07$). With a larger sample size, these changes would have most likely been significant. As such, trends in the data from the MOCA are very similar to the 3D MOT results. This supports the predictive validity of 3D MOT in the assessment of cognitive change in open-heart surgery patients.

4.3. Implications

Taken together, the findings indicate that 3D MOT may be usable to detect post-operative changes in cognition. In addition to replication of the present study, the effects of 3D MOT training after heart surgery need to be tested. Other studies have demonstrated the effectiveness and transferability of such training in elderly patient populations [11,12]. If 3D MOT can similarly improve cognitive function in open-heart patients, it may be used to speed up recovery times for patients and restore basic functions of daily living quicker. In addition, the 3D MOT software is inexpensive and readily available on handheld devices (e.g., tablets). As such, patients may be able to use the software independently and remotely. Future research is needed to test the effectiveness of 3D MOT in restoring post-surgical cognition after open-heart surgery.

4.4. Limitations

There are several limitations that apply to the present pilot study. The study was observational in nature and did not employ a control group. With the findings of the present study, an important next step will be the use of more rigorous designs (e.g., randomized controlled trials). The cognitive function of the participants was examined only upon admission to the hospital (usually 24 to 48 hours before surgery). More information on cognitive changes could be gained by testing the subjects over several time points prior to surgery. The sample size of this pilot study was relatively small, yet some important statistical trends in the data were detected. Because of the voluntary nature of participation and the descriptive design, potential bias in the sampling cannot be excluded. Further, all patients were treated at the same hospital. Other hospitals may handle pre-, intra-, and postoperative care differently, which may impact the findings and hence the

generalizability of the present study. Lastly, the participants recruited for the study all underwent open-heart surgery, but the type of surgery differed (i.e., AVR, CABG, or both). Future studies may control for type of surgery and other intraoperative factors (e.g., length of surgery, type of anesthetic).

5. Conclusions

Post-surgical cognitive changes in heart surgery patients were detected using 3D MOT. Future research may explore whether 3D MOT is usable for the retraining of cognition after heart surgery.

Conflict of interest

None to report.

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